

Solution:

1

Solⁿ is the homogenous mixture of 2 or more component.

Solution = Solute + Solvent
 ↓ ↓
 less in amt. excess in amt.
 (moles) (in terms of moles)

180 g Urea + 90 g water
 → 3 mol → 5 mol
 ↳ solvent

→ 300 g urea + 684 g Sucrose + ~~180~~ 180 g water
 → 5 mol → 2 mol 10 mol

Sol ⁿ component	Solute	Solvent
2	1	1
3	2	1
4	3	1

always = 1

• Nature of solⁿ is same as that of solvent.

Solⁿ

- ↳ solid
- ↳ liquid
- ↳ Gas

depend upon nature of solvent

Solⁿ liq + liq → water + alcohol
 gas + gas → air
 solid + solid → alloys
 liq. + gas → cold drink
 liq. + solid → NaCl + water

Concentration Terms :-

1. Molarity (M)
2. Normality (N)
3. Molality (m)
4. Formality (F)
5. Strength (S)
6. Mole fraction
7. %
 - w/w %
 - ~~v~~ v/v %
 - w/v %

8. PPM

1. Molarity (M) :- no. of moles of solute +nt in 1 L solⁿ.

$$M = \frac{n_{\text{solute}}}{V_{\text{sol}^n} (L)}$$

$$M = \frac{n}{V}$$

$$\text{Unit} = \frac{\text{mol}}{L^{-1}} \quad M$$

2. Normality (N) :- It is no. of g. eq. of solute +nt in 1 L solⁿ

$$N = \frac{\text{no. of g. eq. of solute}}{V_{\text{sol}^n} (L)}$$

$$\text{Unit} = \frac{\text{eq}}{L^{-1}} \quad N$$

$$\text{no. of g. eq.} = \frac{wt}{\text{Eq wt}} = \frac{W}{E} = \frac{W}{M/n\text{-factor}} = \frac{W \times n\text{-factor}}{M} = n \times n\text{-factor}$$

$$\text{no. of g. eq.} = N \times V = \frac{W}{E} = n \times n\text{-factor}$$

$$N = \frac{n \times n\text{-factor}}{V_{\text{sol}^n} (L)}$$

$$N = M \times n\text{-factor}$$

Note: First calculate Molarity then calculate Normality

3. Molality: It is moles of solute +nt in 1 kg solvent.

$$m = \frac{\text{moles of solute}}{\text{wt. of solvent (in kg)}}$$

$$m = \frac{n_{\text{solute}}}{W_{\text{solvent}} (kg)}$$

It does not depend upon Temp.

4. Formality (F): It is no. of formula mass in 1L solⁿ
 → defined for ionic compd.

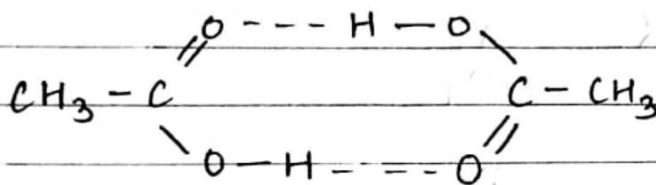
$$F = \frac{\text{Molute}}{(\text{Formula mass}) \times V_{\text{sol}^n} (\text{L})}$$

$$20 \text{ g NaOH in 2 L} \rightarrow F_w = 40 \quad M = \frac{20/40}{2} = 1/4$$

$$M_w = 40$$

$$F = \frac{20}{40 \times 2} = 1/4$$

$$\text{When } F_w = M_w \quad F = M$$



$$F_w = 120$$

$$6 \text{ g CH}_3\text{COOH in 1 L sol}^n \quad M = ? \quad F = ?$$

$$M = \frac{6/60}{1} = 0.1 \text{ M}$$

$$N = \frac{6}{120 \times 1} = 0.05$$

Que₁ 44.8 g benzoic acid is cont in 2 L solⁿ Find Molarity & Formality?

$$\text{C}_6\text{H}_5\text{COOH} = 7 \times 12 + 2 \times 16 + 6$$

$$= 84 + 32 + 6 = 84 + 38 = 122$$

$$M = \frac{\frac{44.8}{122} \times 10}{2} = \frac{2}{10 \times 2} = 0.1 \text{ M}$$

$$F = \frac{44.8}{122 \times 2} = \frac{2}{1 \times 2 \times 10} = 0.1$$

Strength : wt. of solute in 1 L solⁿ

$$S = \frac{\text{wt. of solute}}{\text{Vol}^n (\text{L})} \quad \text{Unit : g/L}$$

Mole fraction : (X)

A	B	C
n_A	n_B	n_C

$$X_A = \frac{n_A}{n_T}$$

$$X_A + X_B + X_C = 1$$

$$X_B = \frac{n_B}{n_T}$$

$$X_C = \frac{n_C}{n_T}$$

$$n_T = n_A + n_B + n_C$$

Solⁿ
$$X_{\text{solute}} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}}$$

$$X_{\text{solvent}} = \frac{n_{\text{solvent}}}{n_{\text{solute}} + n_{\text{solvent}}}$$

$$X_{\text{solute}} + X_{\text{solvent}} = 1$$

$$\frac{X_A}{X_B} = \frac{n_A}{n_B}$$

7.
$$\% \text{ w/w} = \frac{\text{wt. of solute}}{\text{wt. of sol}^n} \times 100$$

$$20\% \text{ w/w Urea} = 20 \text{ g urea in } 100 \text{ g sol}^n$$

$$\% \text{ v/v} = \frac{\text{Vol. of solute}}{\text{Vol. of sol}^n} \times 100$$

$$20\% \text{ v/v NaOH} = 20 \text{ ml NaOH in } 100 \text{ ml sol}^n$$

$$\% \text{ w/v} = \frac{\text{wt. of solute (g)}}{\text{Vol. of sol}^n (\text{ml})} \times 100$$

$$40\% \text{ w/v H}_2\text{SO}_4 \text{ sol}^n = 40 \text{ gm H}_2\text{SO}_4 \text{ in } 100 \text{ ml sol}^n$$

Ques 20% by wt NaOH soln Find molality?

20 gm NaOH + 80 gm solvent in 100 gm soln

$$\text{molality} = \frac{20/40}{100/1000} = \frac{1/2}{1/10} = \frac{10}{2} = 5m$$

Ques 20% w/v NaOH soln find Molarity?

20 gm NaOH + 80 ml solvent in 100 ml soln

$$\text{Molarity} = \frac{20/40}{100/1000} = \frac{1/2}{1/10} = 5M$$

PPM : Parts Per million
very dilute soln

$$\text{ppm (by wt)} = \frac{\text{wt. of solute}}{\text{wt. of solvent soln}} \times 10^6$$

$$\text{PPM (by Vol.)} = \frac{\text{Vol. of solute}}{\text{Vol. of soln}} \times 10^6$$

ppb : Parts Per Billion

$$\text{ppb (by wt)} = \frac{\text{wt. of solute}}{\text{wt. of soln}} \times 10^9$$

AIPMT

Ques How many gram of dibasic acid (Mol. wt. = 200) should be present in 100 ml of aq. soln to give 0.1N soln?

$$N = \frac{W/M_w \times n\text{-factor}}{V_{\text{soln}}}$$

$$N \times V \times M_w = W \times n\text{-factor}$$

$$0.1 \times 100 \times 200 = W \times 2$$

AIPMT-95+92

Ques which of the following concn term is independent of Temp.?

① Molarity ② Molality ③ Normality ④ Formality

Ques Calculate mass of sugar & water req.^d to prepare 250 gm of 20% w/w solⁿ.

~~20 gm sugar + 180 in 100 gm solⁿ~~

$$20 = \frac{Wt \text{ solute}}{250} \times 100$$

$$\frac{250}{5} = W \text{ solute} = 50 \text{ gm}$$

$$W \text{ solvent} = 250 - 50 = 200 \text{ gm}$$

AIPMT-07

Ques Conc. of sulphuric acid is 98% H_2SO_4 by mass & has a density of 1.8 g L^{-1} . Vol of acid req.^d to make one litre of 0.1 M H_2SO_4 solⁿ

① 11.10 ml ② 16.65 ml ③ 22.20 ml ④ 5.55 ml

98 gm H_2SO_4 + 180 in 100 gm solⁿ

$$D = \frac{M}{V}$$

$$1.8 = \frac{98}{V}$$

$$V = \frac{980}{1.8} = 5$$

$$0.1 \text{ M} = \frac{n_{H_2SO_4}}{1} \Rightarrow n_{H_2SO_4} = 0.1 \text{ mol}$$

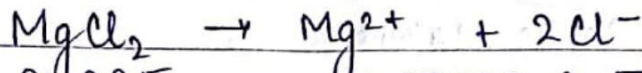
$$V_{H_2SO_4} = 0.1 \times 22.4$$
$$= 2.24$$

Ques Calculate Molarity of 400g pure water if $d_{H_2O} = 1g/ml$

$$\cancel{V = d_{H_2O}} \quad V = \frac{400}{1} = 400 \text{ ml}$$

$$M = \frac{400/18 \times 1000}{400} = \frac{400 \times 1000}{400 \times 18} = 55.5$$

* Ques Calculate Molarity of a solⁿ of $MgCl_2$ if 400ml $MgCl_2$ have 3.011×10^{22} chloride ions



$$0.025 \quad 0.025 \quad 0.05 \text{ mol}$$

$$M = \frac{0.05 \times 10^{-3} \times 1000}{400} =$$

AIPMT-05
JAS

Ques Calc. mole fraction of solute in one molal aq. solⁿ
1 molal $\rightarrow$ 1 mol of solute + nt in 1kg solvent

$$n_{H_2O} = \frac{1000}{18} = 55.5$$

$$X_{\text{solute}} = \frac{1}{1 + 55.5} = \frac{1}{56.5} = \frac{10}{565} = 0.0177$$

NCERT

Ques Calc. mass of urea reqd in making 5kg 0.25 m aq. solⁿ

$$m = \frac{\text{moles of solute}}{\text{wt (kg)}} = \frac{0.25 \times 5}{\text{wt}}$$

$$0.25 \times 5 \times 1000 = w$$

$$w = \frac{1500 \times 5}{1000} = 7500 \text{ gm.}$$

$$39 + 127 = 166$$

Ques Calculate ① Molarity ② Molality ③ Mole fraction of KI if density 20% w/w aq. KI is 1.202 g/ml

$$\textcircled{1} M = \frac{n}{V} = \frac{20/166 \times 1000}{83.3} \quad \begin{array}{l} 20 \text{ gm KI + nt in } 100 \text{ gm sol}^n \\ \rho = 1.202 \end{array}$$

$$= \frac{20 \times 1000}{166 \times 83.3} = 1.44 \quad V = \frac{M}{\rho} = \frac{100}{1.202}$$

$$\textcircled{2} m = \frac{20/166 \times 1000}{80} = \frac{20 \times 1000}{4 \times 80 \times 166} = 1.50 = \frac{10000}{120.2} \text{ ml}$$

$$\textcircled{3} X_{\text{solute}} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}} \quad n_{\text{solute}} = \frac{20}{166} = 0.120$$

$$= \frac{0.12}{0.12 + 4.44} = 0.026 \quad n_{\text{solvent}} = \frac{80}{18} = 4.44$$

Ques Eq. moles of urea & water are mixed to prepare an aq. solⁿ what is w/w % of urea & also find Molality.

$$W/W \% = \frac{1 \times 60 \times 10^5}{2 \times 78} \times 100 = \frac{500}{13} \%$$

$$m = \frac{n \times 1000}{n \times 18} = \frac{1000}{18} = 55.5 \text{ m}$$

Ques Calc. Molarity & molality of 25% w/w H₂SO₄ solⁿ if sp. gravity of solⁿ is 1.2 g/ml

25 gm H₂SO₄ + nt in 100 gm solⁿ

$$M = \frac{25/98 \times 1000}{\frac{100}{1.2}} = \frac{25 \times 1.2 \times 1000}{98 \times 100}$$

$$= \frac{25 \times 12 \times 100}{100 \times 98}$$

$$M = \frac{300}{98} \approx 3M$$

$$\text{molality} = \frac{\frac{25}{98} \times 1000}{75} = \frac{25 \times 1000}{98 \times 75} = \frac{1000}{294} = 3.4m$$

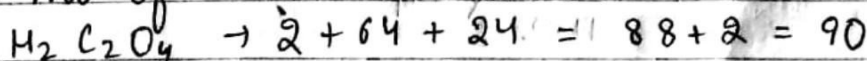
AI PMT

Ques 25.3 g of Na_2CO_3 is dissolved in enough water to make 250 ml of solⁿ. If Na_2CO_3 dissociates completely then molar concⁿ of Na^+ & CO_3^{2-} ions are respectively (Given Mw. of $\text{Na}_2\text{CO}_3 = 106 \text{ g/mol}$)

- (a) 0.955 M & 1.910 M $\text{Na}_2\text{CO}_3 \rightarrow 2\text{Na}^+ + \text{CO}_3^{2-}$
- (b) 1.910 M & 0.955 M c
- (c) 1.90 M & 1.910 M 10 2c c
- (d) 0.477 M & 0.477 M

$$C = \frac{25.3}{106} \times \frac{1000}{250} = \frac{25.3 \times 1000}{106 \times 250} = \frac{50.6}{53}$$

Ques Calculate amt. of oxalic acid req^d to obtain 250 ml of semimolar solⁿ?



$$\frac{1}{2} = \frac{W/90 \times 1000}{250}$$

$$\frac{1}{2} = \frac{W \times 4}{90 \times 45}$$

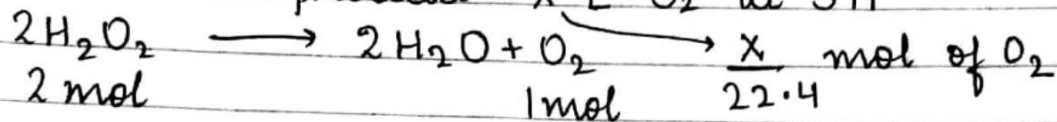
$$\frac{45}{4} = W = 11.25 \text{ gm}$$

Volume Strength of H_2O_2 :- The volume of O_2 produced by decomposition of unit volume of H_2O_2 at NTP/STP

Relation b/w vol. strength of H_2O_2 , M, N & W/V % :-
let vol strength of H_2O_2 = x

Let vol. strength of H_2O_2 is 'x' V

→ 1 L H_2O_2 will produce 'X' L O_2 at STP



1 mol of O_2 is obtained by decomposition of = 2 mol H_2O_2
 $\times$ mol

$$\frac{x}{22.4} \text{ mol} \rightarrow \frac{x}{22.4} \times 2$$

$$= \frac{x}{11.2} \text{ mol of } H_2O_2$$

① Molarity = $\frac{\text{no. of moles}}{\text{Vol (L)}}$

$$M = \frac{\frac{x}{11.2}}{1} = \frac{x}{11.2} \text{ mol l}^{-1}$$

$$M = \frac{X}{11.2}$$

② density $M = \frac{n}{V} = \frac{w}{M_w \times V} = \frac{d}{M_w}$

$$d = M \times M_w$$

$$d = \frac{x}{11.2} \times 34$$

③ Normality $N = M \times n\text{-factor}$
 $= \times$

$$= \frac{X}{11.2} \times 2$$

$$N = \frac{X}{5.6} \quad *$$

⑨ strength : $S = N \times E$

$$S = \frac{X}{5.6} \times 17$$

$$E = \frac{34}{2} = 17$$

$$\textcircled{5} \% \text{ w/v} = \frac{X}{5.6 \times 1000} \times 17 \times 100$$

$$= \frac{X}{5.6} \times \frac{17}{10}$$

Ques Find vol. strength of 1.5 N H_2O_2 solⁿ?

$$N = \frac{X}{5.6}$$

$$1.5 \times 5.6 = X = 8.40 = 8.4 \text{ L}$$

Ques Calculate strength of 44.8 V H_2O_2 solⁿ in terms of ① Molarity ② Normality ③ Strength ④ % w/v

$$\textcircled{1} \text{ Molarity} = \frac{X}{11.2} = \frac{44.8}{11.2} = 4$$

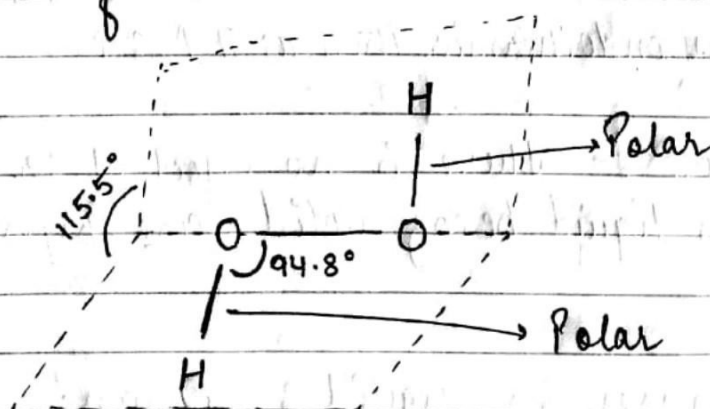
$$\textcircled{2} \text{ Normality} = \frac{X}{5.6} = \frac{44.8}{5.6} = 8$$

$$\textcircled{3} \text{ Strength} = N \times E$$

$$= \frac{X}{5.6} \times 17 \text{ g/L} = 8 \times 17 = 136 \text{ g/L}$$

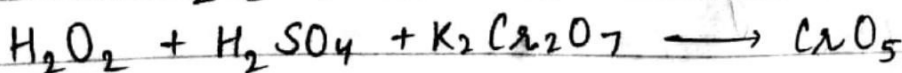
$$\textcircled{4} \% \text{ w/v} = \frac{X}{5.6} \times \frac{17}{10} = \frac{136}{10} = 13.6$$

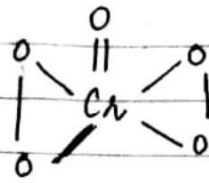
Structure of H_2O_2 :-



Test of H_2O_2

↳ H_2O_2 reacts with Titanic acid $\text{Ti}(\text{OH})_4$





Ca^{2+}
Butterfly str.

Solubility :- The max^m amt. of solute dissolved in a specified amt of solvent is at spec const. Temp.

Solubility depends upon

1. Nature of solute
2. Nature of solvent
3. Temp.
4. Pressure

Solubility Principle :: Like dissolve like

Solubility of solids in liquid : Polar solute are soluble in polar solvent & non-polar solute are soluble in non-polar solvent.

Solute + Solvent $\xrightleftharpoons[\text{crystallisation}]{\text{dissolution}}$ Solution

Effect of Temp. :-

If dissolution process is exothermic then as $T \uparrow$ Solubility $\downarrow$
If dissolution process is endothermic then as $T \uparrow$ $S \uparrow$

Effect of Pressure :- There is no effect of Pr. on solubility of solid in liquid becoz solid are highly incompressible.

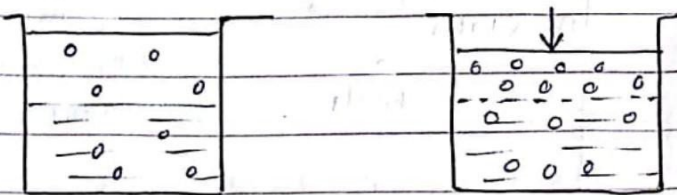
Solubility of gases in liquid :- Depends upon Nature of Gas :

eg NH_3 & HCl are highly soluble then O_2 , N_2 & He are slightly soluble.

Temperature :-

Dissolution of gas in liquid is an endothermic process like condensation ($\text{Gas} \rightarrow \text{Liq.}$) as $\text{Temp} \uparrow$ $\text{Sol.} \downarrow$
 eg: aquatic animals remains in cold water
 Reason: Gas dissolution $\uparrow$

Pressure: $P_r. \uparrow$ $\text{Sol.} \uparrow$



Henry's Law: It states that at constant temp. solubility of a gas in liquid is directly proportional to pressure of gas.
 OR

Mole fraction of a gas in solⁿ is directly proportional to Partial Pressure of the gas over solⁿ.
 OR

The partial Pressure of the gas in vapour Phase (P) is proportional to mole fraction of the gas in solⁿ.
 $P \propto X$

$$P = K_H X$$

$K_H \rightarrow$ Henry's const

$$X \propto \frac{1}{K_H}$$

$T \uparrow$ $\text{Sol.} \downarrow$ $X \downarrow$ $K_H \uparrow$

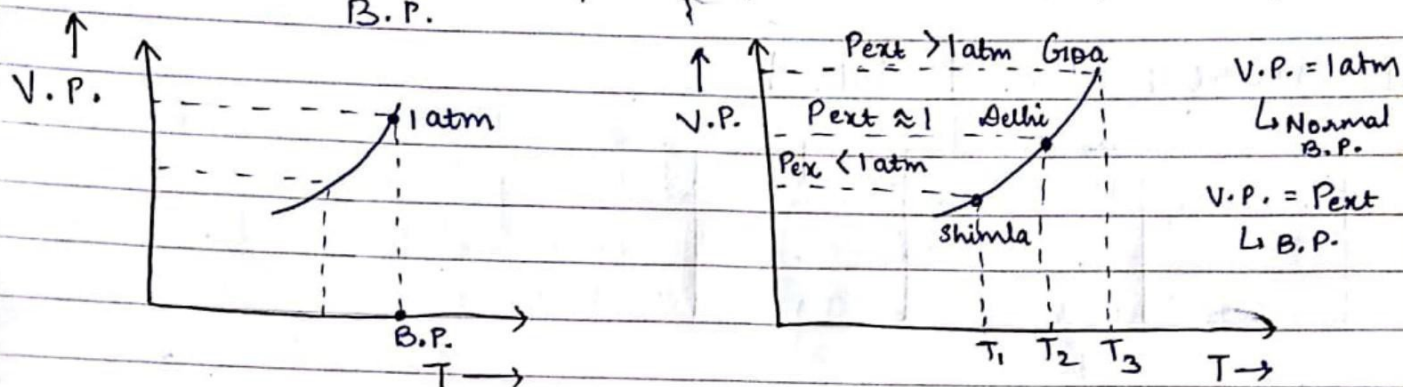
The Pressure exerted by vapour of liquid on its surface when liquid & vapours are in eq^m.



Factors affecting Vapour Pressure :-

1. Nature of liq. :- More vol volatile liq. $VP \uparrow$
2. Temp. :-

Boiling Point: The temp. at which V.P. becomes equal to atmospheric press. Or ext. pr. is e/a



$P_{ext} \uparrow$
 $V.P. \uparrow$
 $B.P. \uparrow$

$P_{ext} \downarrow$
 $V.P. \downarrow$
 $B.P. \downarrow$

Cooking time $\downarrow$ Cooking time $\uparrow$

V.P. of a liquid - liquid Solutions :- (i.e. both liq. are volatile)

Raoult's Law :- The V.P. of a liquid is directly proportional to mole fraction of liq. in soln.

$$\begin{array}{cc} \boxed{\frac{P_A^0}{A}} + \boxed{\frac{P_B^0}{B}} = \boxed{\frac{P_A^0 + P_B^0}{A+B}} \\ P_A \propto X_A & P_B \propto X_B \\ P_A = P_A^0 X_A & P_B = P_B^0 X_B \end{array}$$

$P_A \rightarrow$ Partial V.P. of A
 $P_B \rightarrow$ Partial " " B
 $P_A^0 \rightarrow$ V.P. of Pure liq. A
 $P_B^0 \rightarrow$ " " " " B

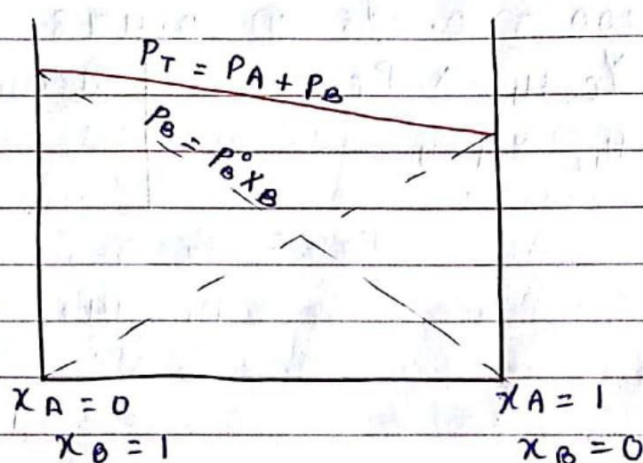
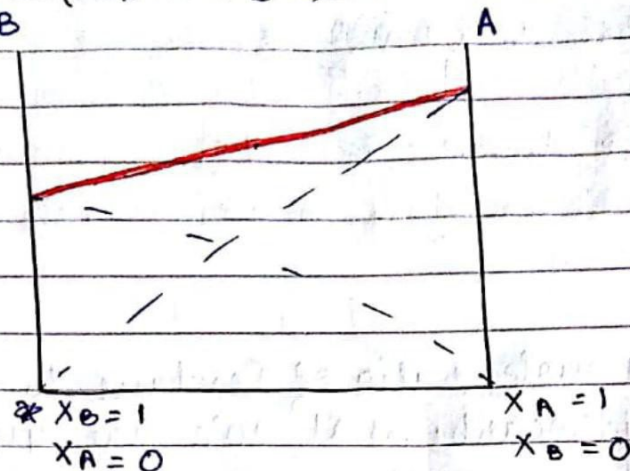
Total Pr. = Sum of Partial Pr. of each compd.

$$P_T = P_A + P_B$$

$$P_T = P_A^0 X_A + P_B^0 X_B$$

$$P_T = P_A^0 X_A + P_B^0 (1 - X_A)$$

$$P_T = (P_A^\circ - P_B^\circ) X_A + P_B^\circ \quad A > B$$



Mole fraction of A & B in vapour phase :-
Dalton's Law

Partial Pr. = mole fraction $\times P_T$

$$P_A = Y_A \times P_T \quad P_B = Y_B \times P_T$$

$$P_A^\circ X_A = Y_A \times P_T \quad P_B^\circ X_B = Y_B \times P_T$$

$Y_A \rightarrow$ mole fraction of A in vapour phase

$Y_B \rightarrow$ mole fraction of B in vapour phase

Ques The V.P. of a liquid P & Q are 80 & 60 torr resp.
The total V.P. of solⁿ obtain by 3 moles of P & 2 moles Q would be.

- ① 140 torr ② 20 torr ③ 68 torr ④ 72 torr

$$\begin{aligned}
 P_T &= P_A^\circ X_A + P_B^\circ X_B \\
 &= \frac{88}{5} \times \frac{3}{5} + \frac{60}{11} \times \frac{2}{5} \\
 &= 48 + 22 = 70
 \end{aligned}$$

Ques A solⁿ has 1:4 mole ratio of Pentane to hexane. The V.P. of pure hydrocarbons at 20°C are 440 mm of Hg for pentane & 120 mm of Hg for hexane. The mole fraction of Pentane in vapour phase would be -

- ① 0.549 ② 0.200 ③ 0.786 ④ 0.478

$$\begin{aligned}
 P_{C_5H_{12}}^\circ X_{C_5H_{12}} &= Y_{C_5H_{12}} \times P_T & \left| \frac{n_{C_5H_{12}}}{n_{C_6H_{14}}} = \frac{1}{4} \right. \\
 \frac{440}{88} \times \frac{1}{5} &= Y_{C_5H_{12}} \times 184
 \end{aligned}$$

$$Y_{C_5H_{12}} = \frac{88}{184} \approx 0.478$$

$$Y_{C_5H_{12}} = 0.478$$

$$\begin{aligned}
 P_T &= 440 \times \frac{1}{5} + 120 \times \frac{4}{5} \\
 &= 88 + 96 \\
 &= 184
 \end{aligned}$$

For solid - liq. solⁿ :-

A → solvent (liquid) → volatile

B → solute (solid) → non-volatile

⇒ V.P. of liquid in pure state → P_A°

V.P. of solid in pure state → $P_B^\circ = 0$

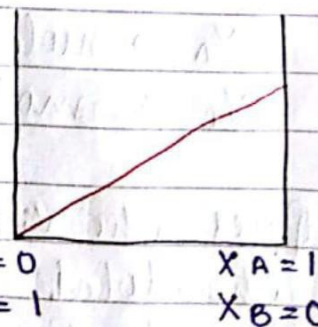
$$P_T = P_A + P_B$$

$$P_T = P_A^\circ X_A + P_B^\circ X_B \rightarrow 0$$

$$P_T = P_A^\circ X_A$$

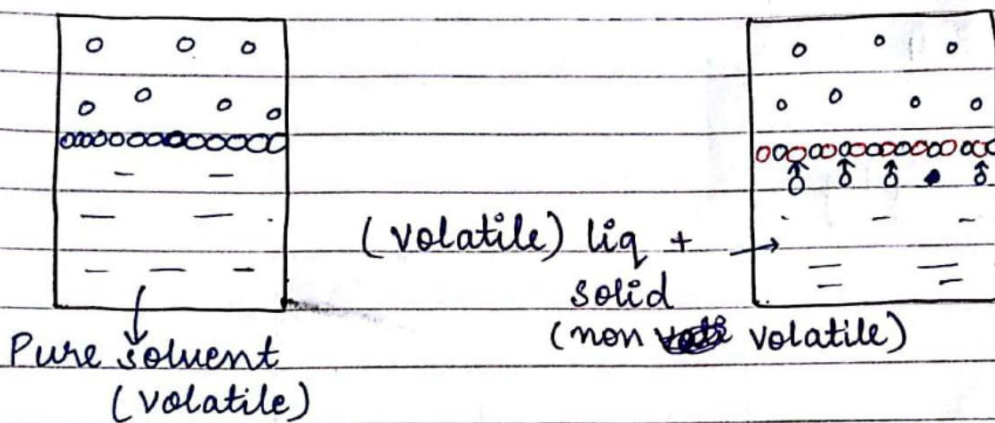
$$P_T = P_A^\circ \frac{N}{n+N}$$

N → moles of solvent
n → moles of solute



→ In solid-liquid solⁿ, v.p. of solⁿ is directly proportional to mole fraction of solvent (i.e. amt. of volatile component)

When a non-volatile solute is added to volatile solvent then its vapour Pressure decrease because some surface is covered by non-volatile solute so less vapours are formed hence V.P. ↓ ses.



n → mole of solute (non volatile)

N → moles of solvent (volatile)

P° → V.P. of pure solvent

$$P_T = P_A^\circ \times A + P_B^\circ \times B^\circ$$

$$P_T = P^\circ \times \frac{N}{n+N}$$

$$\frac{P_T}{P^\circ} = \frac{N}{n+N}$$

$$\frac{P^\circ}{P_T} = \frac{n+N}{N}$$

$$\frac{P^\circ}{P_T} = \frac{n}{N} + 1$$

$$\frac{P^\circ - P_T}{P_T} = \frac{n}{N}$$

→ to find out wt. & molecular wt. of solute / solvent

$$P^\circ - P_T = \Delta P \rightarrow \text{relative lowering V.P.}$$

Ques The vapour pressure at a given temp. of an ideal solⁿ containing 0.2 mole of a non-volatile solute & 0.8 mole of solvent is 60 mm of Hg. Find vapour pressure of pure solvent at same T.

- (a) 150 mm of Hg (b) 60 mm of Hg (c) 75 mm of Hg (d) 120 mm of Hg

$$\frac{P^{\circ} - P_T}{P_T} = \frac{n}{N}$$

$$\frac{P^{\circ} - 60}{60} = \frac{0.2}{0.8}$$

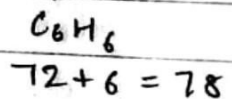
$$P^{\circ} - 60 = 15$$

$$P^{\circ} = 75 \text{ mm of Hg}$$

Ques Vapour Pressure of benzene at 30°C is 121.8 mm of Hg when 15 g of non-volatile solute is dissolved in 250 g of benzene its Vapour Pr. is reduced to 120.2 mm of Hg. Find Mw of solute.

- (1) 356.2 (2) 456.3 (3) 350.5 (4) 656.7

$$\frac{121.8 - 120.2}{120.2} = \frac{15}{\text{Mw} \times \frac{250}{78}}$$



$$\frac{1.6}{120.2} = \frac{15 \times 78}{\text{Mw} \times 250}$$

$$\text{Mw} = \frac{234 \times 120.2}{50 \times 1.6}$$

$$= \frac{234 \times 120.2}{80} = 351.0$$

Ques If v.p. of a solⁿ a Glucose in water is 750 mm at 373K. Find out (1) mole fraction of Glucose (2) Molality of solⁿ

$$P_T = P_A^0 X_A + P_B^0 X_B$$

$$750 = 760 X_A$$

$$X_A = \frac{750}{760} = \frac{75}{76}$$

$$X_B = 1 - X_A = 1 - \frac{75}{76} = \frac{1}{76}$$

$$\textcircled{2} m = \frac{\text{moles}}{\text{wt (kg)}} = \frac{\frac{1}{76}}{\frac{75}{76} \times \frac{18}{1000}} = \frac{1}{75} \times \frac{1000}{18} = \frac{1}{75} \times 55.5$$

Que V.P. of pure water is 112.5 mm of Hg at 300K. Find out the V.P. of 1 molal solⁿ of a non-volatile solute in it?

1 mol of solute + nt 1 kg solvent

$$\rightarrow 1000g \quad n = \frac{1000}{18} = 55.5$$

$$P_T = P_A^0 X_A + P_B^0 X_B$$

$$= 112.5 \times \frac{55.5}{1 + 55.5}$$

$$P_T = 112.5 \times \frac{55.5}{56.5}$$

Que Calc. the wt. of non-volatile solute, should be dissolved in 78g benzene to reduce its V.P. to 80%?

(Given $M_w^{\text{of solute}} = 60 \text{ g/mol}$)

$$\frac{100 - 80}{80} = \frac{w/60}{78/78}$$

$$\frac{20}{80} = \frac{w}{60}$$

$$w = \frac{60 \times 20}{80} = 15 \text{ gm}$$

$$P^0 = 100$$

$$P_T = 80$$

Ques The V.P. of liq. A & B are in ratio of $P_A^0 : P_B^0 = 1:2$ at certain temp., the mole ratio of A & B in liq. solⁿ is 1:2, $\frac{1}{5}$ what is mole fraction of A in vapour phase?

$$\frac{n_A}{n_B} = \frac{1}{2} = \frac{X_A}{X_B}$$

$$\frac{P_A^0 X_A}{P_B^0 X_B} = \frac{Y_A \times P_T}{Y_B \times P_T}$$

$$\frac{1}{2} \times \frac{1}{2} = \frac{Y_A}{Y_B} = \frac{1}{4} = \frac{n'_A}{n'_B}$$

$$Y_A = \frac{1}{5}$$

Ques 2 liq. A & B form an ideal solⁿ at 30°C. The V.P. of solⁿ = 2 mol of A & 3 mol of B is 600 mm of Hg. But if 1 mole of A & 4 moles of B are mixed then the V.P. of solⁿ formed is 620 mm of Hg. what is V.P. of pure A & pure B?

$$P_T = P_A^0 X_A + P_B^0 X_B$$

$$600 = P_A^0 \cdot \frac{2}{5} + P_B^0 \left(\frac{3}{5}\right) \quad \text{--- (1)}$$

$$620 = P_A^0 \cdot \frac{1}{5} + P_B^0 \cdot \frac{4}{5} \quad \text{--- (2)}$$

$$\frac{30 \cdot 600}{31} = \frac{\frac{1}{5} (2P_A^0 + 3P_B^0)}{\frac{1}{5} (P_A^0 + 4P_B^0)}$$

$$\frac{30}{31} = \frac{2P_A^0 + 3P_B^0}{P_A^0 + 4P_B^0}$$

$$30P_A^0 + 120P_B^0 = 62P_A^0 + 93P_B^0$$

$$27P_B^0 = 32P_A^0$$

$$\frac{27}{32} = \frac{P_A^0}{P_B^0}$$

Ques The V.P. of mix of 2 volatile compound A & B is 0 P = 254 - 119x mm of Hg (x = mole fraction of B in liq. solⁿ) Find P_A^0 & P_B^0

$$P_T = 254 - 119x$$

$$= P_A^0 X_A + P_B^0 X_B$$

$$P_T = P_A^\circ (1 - X_B) + P_B^\circ X_B$$

$$P_T = P_A^\circ + (P_B^\circ - P_A^\circ) X_B$$

$$P_T = P_A^\circ - (P_A^\circ - P_B^\circ) X_B$$

$$P_A^\circ = 254 \text{ mm of Hg}$$

$$P_A^\circ - P_B^\circ = 119$$

$$254 - 119 = P_B^\circ$$

$$P_B^\circ = 135 \text{ mm of Hg}$$

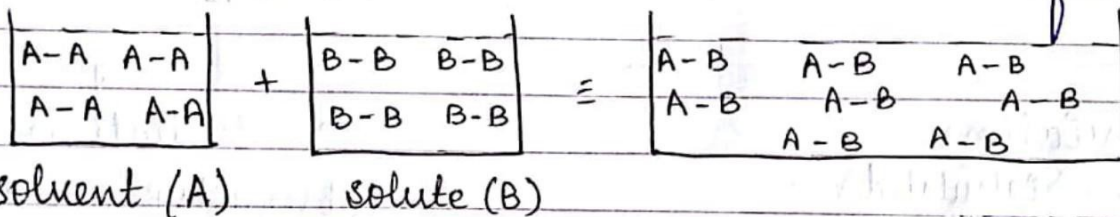
V. Imp
**

Ideal solution :-

Solⁿ which follow Raoult's law at all Temp. & all concⁿ, called ideal solⁿ.

• $\Delta H_{mix} = 0$ i.e. there is no change in heat during formⁿ of solⁿ

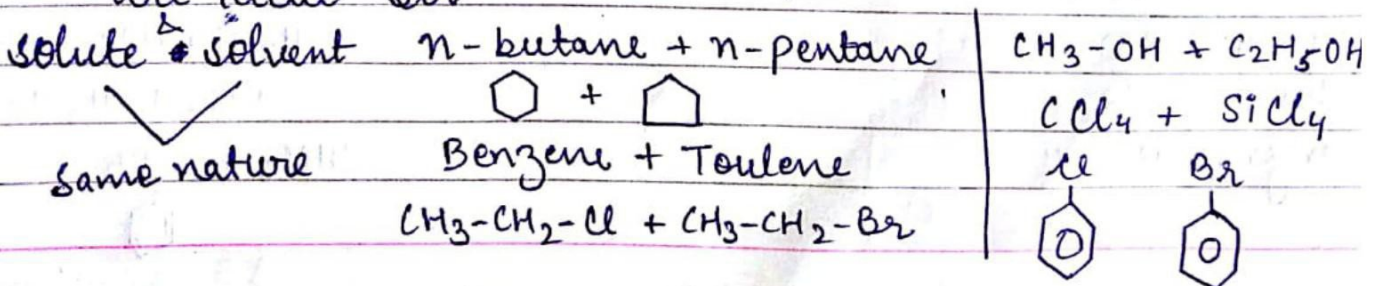
$\Delta V_{mix} = 0$ Vol. of solⁿ = Vol^m of solute + Vol^m of solvent



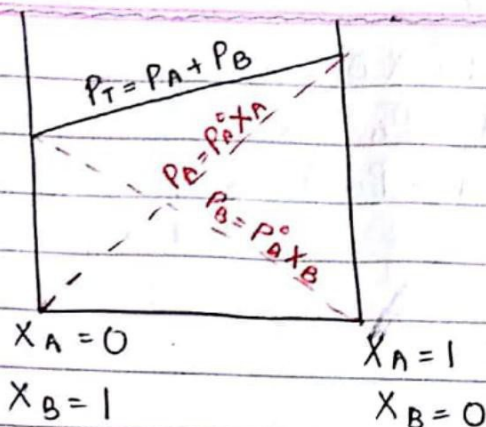
Forces of interaction b/w solute + Forces of interaction b/w solvent = Forces of interaction b/w solute & solvent particles

$A-A + B-B = A-B$

Note: No solⁿ is ideal / perfect solⁿ however solⁿs which have same type of solute & solvent particles are ideal solⁿ



A → solvent
B → solute



$$\Delta S = 0$$

Note: Dilute solⁿ are considered as ideal solⁿ becoz amt. of solute ~~Real solut~~ is very less therefore effect of interaction b/w molecules (A-B) are ~~neg~~ neg

V. Imp

Real solⁿ / Non-Ideal solⁿ :-
→ solⁿ which does not follow Raoult's law.

Observed V.P. (exp.) $\neq$ calculated V.P. (Theo.)

observed B.P. $\neq$ calculated B.P.

+ve Deviation

→ observed V.P. > calculated V.P.

$$\Delta H_{mix} > 0$$

$$\Delta V_{mix} > 0$$

A-A & B-B > A-B

Obs. BP < Calc. BP.

eg CCl₄ & CHCl₃

CCl₄ & C₂H₅OH

Benzene & CCl₄

* C₂H₅OH & CH₃-C(=O)-CH₃

C₂H₅OH & H₂O

Phenol & CS₂

Benzene & acetone

$$\Delta S > 0$$

-ve Deviation

observed VP < calc. V.P.

$$\Delta H_{mix} < 0$$

$$\Delta V_{mix} < 0$$

A-A & B-B < A-B

Obs. B.P. > Cal. B.P.

CHCl₃ + CH₃-C(=O)-CH₃

H₂O & HNO₃

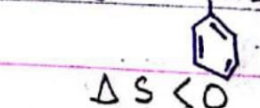
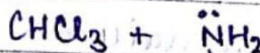
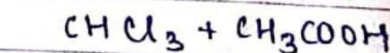
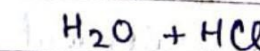
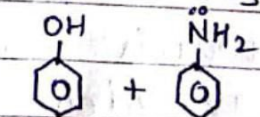
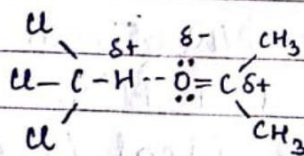
OH + NH₂

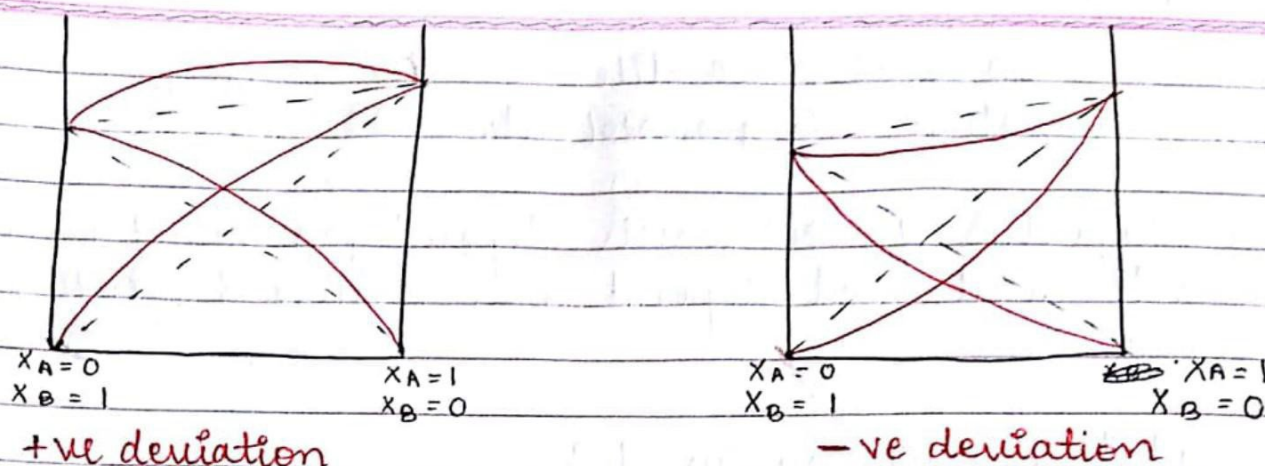
H₂O + HCl

CHCl₃ + CH₃COOH

CHCl₃ + NH₂

$$\Delta S < 0$$





Azeotropic Mixture :- These are constant Boiling Mixture.

Azeotrophs are formed by mixing 2 different liquid in a fixed proportion so that the whole mixture boils at const. boiling temp.

* Azeotrophs can't be separated by fractional distillation.

→ Fractional Distillation :-

Method of separating 2 liquids from their mixture on the basis of their different B.P.

Two types of Azeotropics :-

Min^m Boiling azeotrophs ; Max^m Boiling azeotrophs.

eg : +ve deviation real solⁿ ; -ve deviation real solⁿ.

eg : $C_2H_5OH + H_2O$
95% 5%

eg $H_2O + HNO_3$
32% 68%

B.P. of azeotrophs < B.P. of component

Colligative Properties :-

solvent $\rightarrow$ volatile $\rightarrow A$
 solute $\rightarrow$ non volatile $\rightarrow B$] V.P. $\downarrow$

The properties of solⁿ which depend upon no. of particles in solⁿ and not depend upon nature & identity of solute.

1. Relative lowering in V.P.
2. Elevation in B.P.
3. Depression in Freezing Point
4. Osmotic Pressure

These all 4 properties are connected with V.P.

1. Relative Lowering in V.P. :-

solvent $\rightarrow$ volatile $\rightarrow A \rightarrow N$ moles
 solute $\rightarrow$ non volatile $\rightarrow B \rightarrow n$ -moles] V.P. $\downarrow$

solⁿ

$$P_B^0 = 0$$

Accⁿ to Raoult's law

$$P_T = P_A + P_B$$

$$P_T = P_A^0 X_A + P_B^0 X_B$$

$$P_T = P_A^0 X_A$$

$$P_T = P_A^0 (1 - X_B)$$

$$\frac{P_A^0 - P_T}{P_A^0} = X_B$$

$$\frac{P_A^0 - P_T}{P_A^0} = \frac{n}{n+N}$$

relative lowering of V.P. $\leftarrow \frac{\Delta P}{P_A^0} = \frac{n}{n+N}$

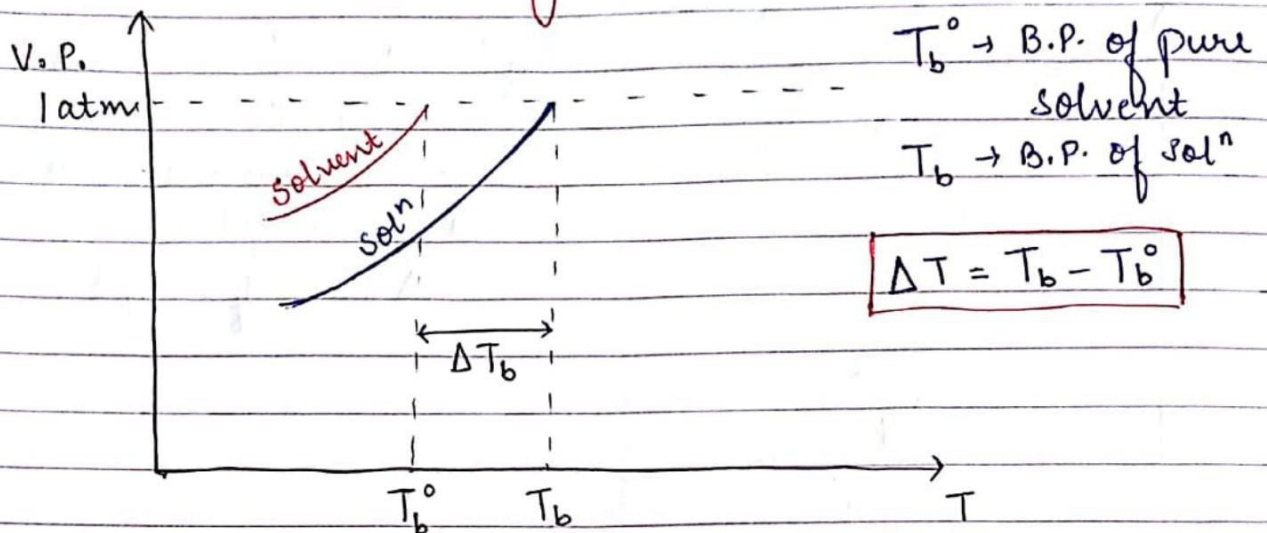
dilute solⁿ

$$n+N \approx N$$

$$\frac{\Delta P}{P_A^0} = \frac{n}{N}$$

$$\frac{\Delta P}{P_A^\circ} = \frac{W_B/M_B}{W_A/M_A}$$

2. Elevation in Boiling Point (ΔT_b) :-



ΔT_b is directly proportional to molality

$$\Delta T_b \propto m$$

$$\Delta T_b = K_b m$$

Unit of $K_b \rightarrow K \text{ kg mol}^{-1}$

$K_b \rightarrow$ Elevation const./
Molal Elevation const
or

Ebullioscopic const.

$$\Delta T_b = K_b \times \frac{W_B/M_B}{W_A/1000}$$

$$m = \frac{n_{\text{solute}}}{W_{\text{solvent (in kg)}}$$

$$\Delta T_b = K_b \times \frac{W_B \times 1000}{M_B \times W_A}$$

To calculate
molar
mass
of solute

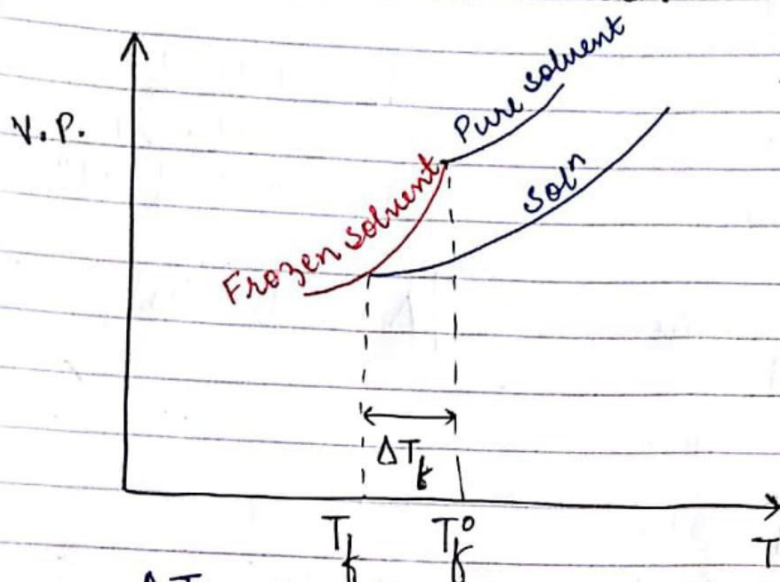
$$M_B = \frac{K_b \times W_B \times 1000}{\Delta T_b \times W_A}$$

$$K_b = \frac{R \times M_A \times T_b^{\circ 2}}{1000 \times \Delta_{\text{vap}} H}$$

$R \rightarrow$ Universal gas const.
 $M_A \rightarrow$ molar mass of solvent
 $T_b^0 \rightarrow$ B.P. of pure solvent
 $\Delta_{\text{vap}} H \rightarrow$ Heat of vap.

3. Depression in Freezing Point (ΔT_f)

Freezing Point $\div$ The temp. at which vapour pressure of solvent in liq. phase is same that of V.P. in solid state.



$T_f^0 \rightarrow$ F.P. of Pure solvent in liq. Phase

$T_f \rightarrow$ F.P. when non-volatile solute is added in it

$$\Delta T_f = T_f^0 - T_f$$

ΔT_f is directly proportional to molality

$$\Delta T_f \propto m$$

$$\Delta T_f = K_f m$$

Unit of $K_f \rightarrow K \text{ kg mol}^{-1}$

$K_f \rightarrow$ Depression const / molal Depression const

(cryo cryoscopic const.)

$$\Delta T_f = K_f \times \frac{W_B / M_B}{W_A / 1000}$$

$$\Delta T_f = K_f \times \frac{W_B \times 1000}{M_B \times W_A}$$

to calculate molar mass of solute $\leftarrow$

$$M_B = \frac{K_f \times W_B \times 1000}{\Delta T_f \times W_A}$$

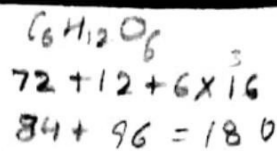
$$K_f = \frac{R \times M_A \times T_f^{02}}{1000 \times \Delta_{\text{fusion}} H}$$

$R \rightarrow$ Universal gas const.

$M_A \rightarrow$ molar mass of solvent

$T_f^0 \rightarrow$ F.P. of pure solvent

$\Delta H_{\text{fusion}} \rightarrow$ heat of fusion



27

NCERT

Ques 36 g of glucose is dissolved in 500 gm water. At what temp. will be water boil? (K_b) water = $0.52 \text{ K kg mol}^{-1}$

$$\begin{aligned} \Delta T_b &= \frac{K_b \times W_B \times 1000}{M_B \times W_A} \\ &= \frac{0.52 \times 36 \times 1000}{180 \times 500} \\ &= \frac{104}{5 \times 100} \end{aligned}$$

$$\Delta T_b = 20.8 \times 10^{-2}$$

$$\begin{aligned} \Delta T_b &= T_b - T_b^\circ \\ \Delta T_b + T_b^\circ &= T_b \\ 0.208 + 373 &= T_b \\ T_b &= 373.208 \end{aligned}$$

$W_B = 36 \text{ g}$
 $W_A = 500 \text{ g}$
 $K_b = 0.52$

Ques The B.P. of Benzene is 353.23 K when 1.80 g non-volatile solute was dissolved in 90 g of Benzene. The B.P. is raised to 354.11 K . Calc. molar mass of solute K_b for benzene = $2.53 \text{ K kg mol}^{-1}$

$$\begin{aligned} M_B &= \frac{K_b \times W_B \times 1000}{\Delta T_b \times W_A} \\ &= \frac{2.53 \times 1.80 \times 1000}{0.88 \times 90} \end{aligned}$$

$$= \frac{2530}{44} = 57.5 \text{ gm}$$

$$\Delta T_b = 354.11 - 353.23 = 0.88$$

$$\begin{array}{r} 354.11 \\ - 353.23 \\ \hline 0.88 \end{array}$$

$$\begin{array}{r} 2530 \\ \div 44 \\ \hline 57.5 \end{array}$$

Ques B.P. of water at 750 mm of Hg is 99.63°C . How much sucrose is to be added to 500 gm of water such that it boils at 100°C

$$\begin{aligned} \Delta T_b &= T_b - T_b^\circ \\ &= 100 - 99.63 = 0.37^\circ \text{C} \end{aligned}$$

$$\Delta T_b = K_b m$$

$$\Delta T_b = K_b \times \frac{W_B / M_B \times 1000}{W_A}$$

$$0.37 = 0.52 \times \frac{W_B \times 1000}{342 \times 500}$$

$$\frac{0.37}{0.52} \times \frac{171}{2} = W_B$$

$$W_B = \frac{63.27}{0.52} = 121.67 \text{ g} \approx 122 \text{ g}$$

$$72 + 8 + 96 = 176$$

Ques Calc. Mass of Ascorbic Acid (Vitamin C, $C_6H_8O_6$) to be dissolved in 75 gm of acetic acid to lower its M.P. by ~~1.5~~ $1.5^\circ C$ ($K_f = 3.9^\circ K \text{ kg mol}^{-1}$)

$$M_B = \frac{K_f \times W_B \times 1000}{\Delta T_f \times W_A}$$

$$W_B = \frac{M_B \times \Delta T_f \times W_A}{K_f \times 1000}$$

$$= \frac{176 \times 1.5 \times 75}{3.9 \times 1000}$$

$$W_B = 5.076 \text{ gm}$$

Ques A solⁿ of sucrose (Molar Mass = 342 g mol^{-1}) has been prepared by dissolving 68.5 gm of sucrose in 1000 gm of water. The F.P. of solⁿ will be - (water $\rightarrow K_f \approx 1.86^\circ K \text{ kg mol}^{-1}$)

- ① $-0.372^\circ C$ ② $-0.520^\circ C$ ③ $+0.372^\circ C$ ④ $-0.570^\circ C$

$$\Delta T_f = \frac{K_f \times W_B \times 1000}{M_B \times W_A}$$

$$= \frac{1.86 \times 68.5 \times 1000}{342 \times 1000}$$

$$\Delta T_f = 0.3725$$

$$\Delta T_f = T_f^0 - T_f$$

$$T_f = T_f^0 - \Delta T_f$$

$$T_f = 273 - 0.3725$$

$$T_f = 272.6275 \text{ K}$$

AIAMT
Ques

A solⁿ of urea (Molar Mass = 56 g mol^{-1}) boils at 100.18°C at atm. pressure. If K_f & K_b for water are $1.86 \text{ K kg mol}^{-1}$ & $0.512 \text{ K kg mol}^{-1}$ resp. The above solⁿ will freeze at.

- (1) -6.54°C (2) 6.54°C (3) 0.654°C (4) -0.654°C

$$\Delta T_f = \frac{K_f \times W_B \times 1000}{M_B \times W_A}$$

$$\Delta T_b = \frac{K_b \times W_B \times 1000}{M_B \times W_A}$$

$$\frac{\Delta T_f}{\Delta T_b} = \frac{K_f}{K_b}$$

$$\frac{\Delta T_b}{100.18 - 100} = \frac{1.86}{0.512}$$

$$\Delta T_b = \frac{1.86 \times 0.18}{0.512} = 0.6539^\circ \text{C}$$

$$\Delta T_b = T_b^0 - T_b$$

$$T_b = T_b^0 - \Delta T_b$$

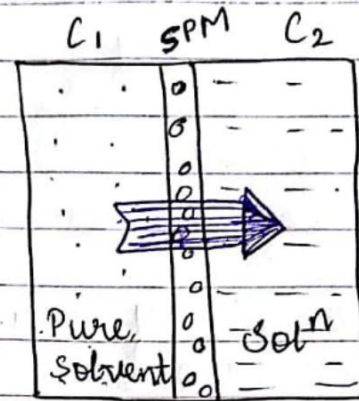
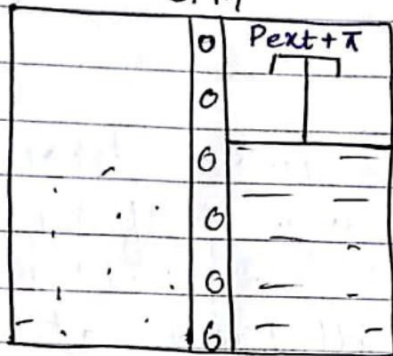
$$T_b = 273 - 0.6539$$

$$T_b = 272.346 \text{ K}$$

$$T_b = -0.6539^\circ \text{C}$$

4. Osmosis :- The flow of liquid from low dilute solⁿ (low concⁿ) to concentrated solⁿ (high concⁿ) through semi permeable membrane is c/a osmosis.

Osmotic Pressure (π) :



The min^m ext. pr. required to apply on solⁿ side to prevent osmosis, is c/a Osmotic pressure (π)

Osmotic pr. $\propto$ conc. of solⁿ

$$\pi \propto C$$

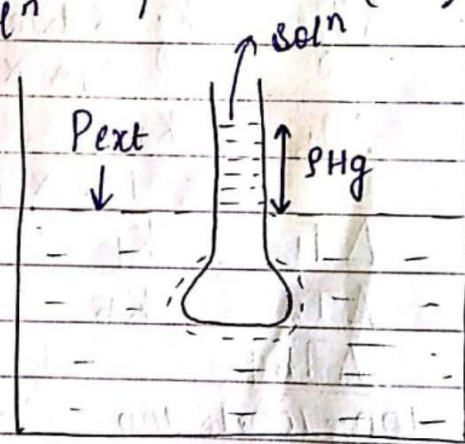
$$\pi = CRT$$

$$\pi = n_B / V RT$$

$$\pi = \frac{W_B / M_B RT}{V}$$

Formula to calculate M_w to solute.

$$M_B = \frac{W_B RT}{\pi V}$$



AIPMT

This method is used to find M.w. of biomolecules like proteins, polymers & other macromolecules becoz they are not stable at high temp.

Type of Solution :-

1. Isotonic solⁿ : Two solⁿ have same osmotic pressure at a given temp. c/a isotonic solⁿ.

when these 2 solⁿ are separated by SPM there is no osmosis.

2. Hypertonic & Hypotonic solⁿ :-
osmotic pr. of blood $\approx 0.9\%$ ($\frac{w}{v}$) of NaCl

Hypertonic solⁿ :-

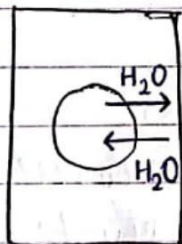
solⁿ which have concⁿ more than 0.9% w/v NaCl

→ when cell is placed in such solⁿ water will flow out of cell and cell would shrink.

Hypotonic solⁿ :-

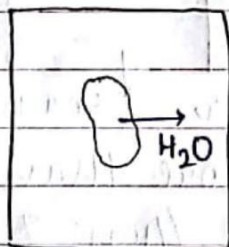
solⁿ which have concⁿ less than 0.9% w/v NaCl.

→ when cell is placed in such solⁿ water will flow inside the cell and cell would swell up.



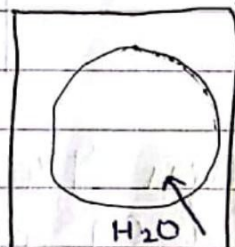
conc. $\approx 0.9\%$ w/v NaCl

→ isotonic



conc. $> 0.9\%$ w/v NaCl

→ hypertonic
cell shrink



concⁿ $< 0.9\%$ w/v NaCl

Hypotonic solⁿ

Edema :- When a person take a lot of salt or salty food water comes out of RBC & water ~~and~~ retention in tissue take place which cause intracellular space resulting in swelling, this is k/n as Edema.

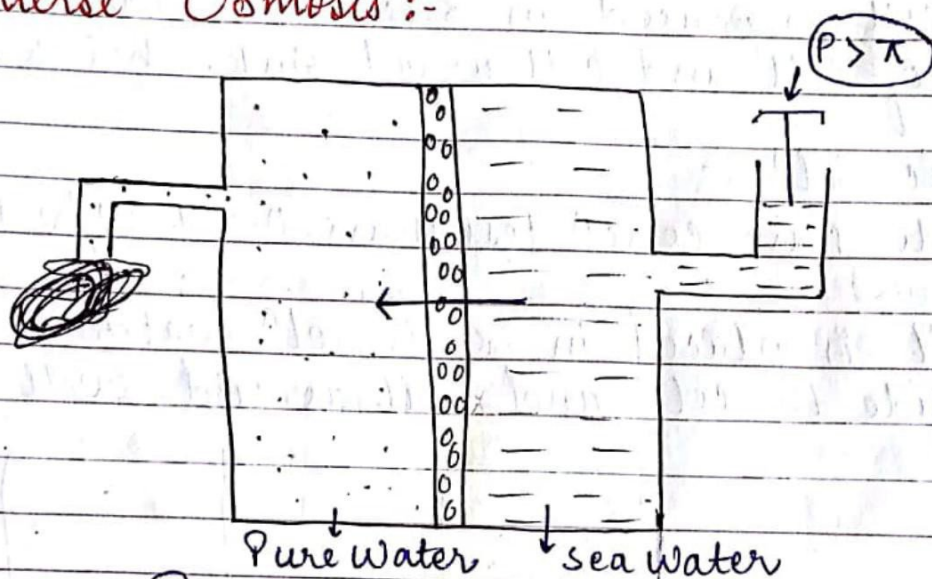
Other Example of Osmosis :-

- Raw mangoes placed in salty water loses water via osmosis & shrivel into pickles.
- Wilted flower revive when placed in water

Isohydric Solutions :-

2 solⁿs which have same H^+ concⁿ i.e. same pH c/a isohydric solⁿ.

Reverse Osmosis :-



When Pressure more than osmotic Pressure is applied on solⁿ side then ϕ molecules flow from solⁿ to solvent. This process is k/n as reverse osmosis.

AIRMT

Que A 5% solⁿ of cane sugar ($M_w = 342$) is isotonic with 1% solⁿ of a substance X. The Mol. wt. of X is -

- ① 34.2 ② 171.2 ③ 68.4 ④ 136.8

$$\pi_1 = \pi_2$$

$$\frac{5}{100} \times \frac{1}{342} \times \frac{RT}{V} = \frac{1}{100} \times \frac{1}{M} \times \frac{RT}{V} \Rightarrow \frac{5}{342} = \frac{1}{M}$$

$$M = \frac{342}{5} = 68.4$$

$$M = 150$$

Que A solⁿ is prepared by 10 g of an org. compd in one litre solⁿ at 27°C osmotic pr. of solⁿ is 1.64 atm. Calculate molar mass of solvent

$$\pi = CRT$$

$$1.64 = \frac{W/M}{V} RT$$

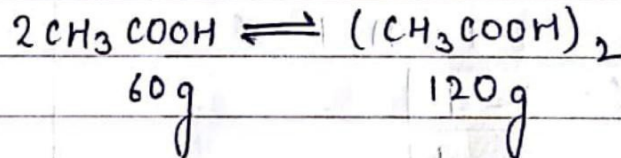
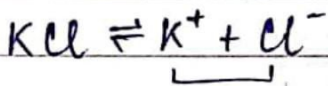
$$M = \frac{W RT}{V \times 1.64}$$

$$= \frac{10}{1} \times \frac{0.082 \times 300 \times 1000}{1.64 \times 1000}$$

$$M = 150$$

Vant Hoff factor (i) :-

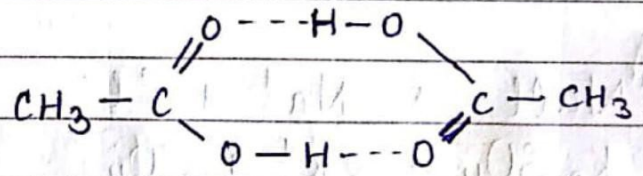
Colligative Properties depends upon no. of particle.



dissociation

Association

$$i = \frac{\text{Observed C.P.}}{\text{Calculated C.P.}}$$



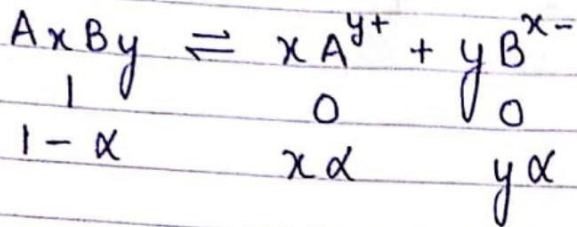
$$i = \frac{\text{normal molar mass}}{\text{abnormal mass}}$$

$$i = \frac{\text{no. of particles after dissociation / association}}{\text{no. of particles before dissociation / association}}$$

Dissociation :-

no. of Particles $\uparrow$
 $i > 1 \Rightarrow \text{Obs. C.P.} > \text{Calc. C.P.}$

normal molar mass $>$ abnormal molar mass



$$\begin{aligned} \text{Total mole after dissociation} &= 1 - \alpha + x\alpha + y\alpha \\ &= 1 - \alpha + (x+y)\alpha \\ &= 1 - \alpha + n\alpha \end{aligned}$$

$$i = \frac{\text{total moles after dissociation}}{\text{total moles before dissociation}} \quad \begin{array}{l} x+y = n \\ n \rightarrow \text{Total dissociat} \\ \text{moles ion} \end{array}$$

$$i = \frac{1 - \alpha + n\alpha}{1}$$

For strong electrolyte
 DOD = 100% $\alpha = 1$

Calculation of DOD $\leftarrow$

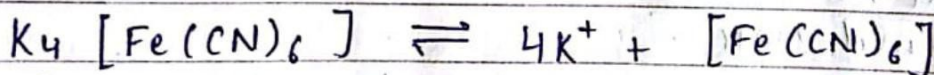
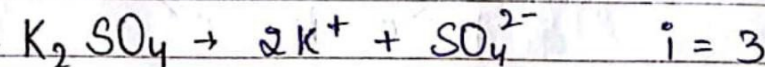
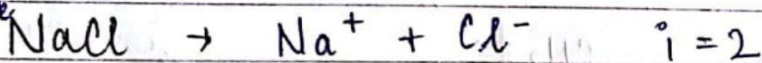
$$i = 1 + (n-1)\alpha$$

$$\alpha = \frac{i-1}{n-1}$$

$$1 = \frac{i-1}{n-1}$$

$$i = n$$

strong electrolyte



(1) If $\alpha = 100\%$ $\alpha = \frac{i-1}{n-1}$

$$i = n$$

$$i = 5$$

(2) If $\alpha = 50\%$ $\frac{50}{100} = \frac{i-1}{5-1}$

$$5-1 = 2i-2$$

$$4 = 2i - 2$$

$$i = 3$$

Observed C.P. = i Calc. C.P.

$$\frac{\Delta P}{P^0} = i \times \frac{n}{N}$$

$$\pi = i CRT$$

$$\Delta T_b = i \times K_b m$$

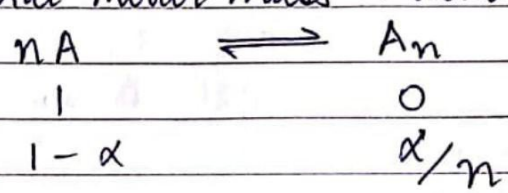
$$\pi = i \frac{n_B}{V} RT$$

$$\Delta T_f = i \times K_f m$$

Association : $i < 1$

obs. C.P. < Calc. C.P.

normal molar mass < abnormal molar mass



Total moles after association = $1 - \alpha + \alpha/n = 1 - (1 - \frac{1}{n})\alpha$

Total moles before association = 1

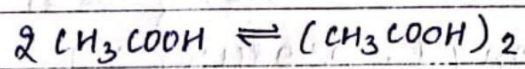
$$i = \frac{1 - (1 - \frac{1}{n})\alpha}{1}$$

$$\alpha = \frac{1 - i}{1 - \frac{1}{n}}$$

DOD = 100%

$$\alpha = 1 = \frac{1 - i}{1 - \frac{1}{n}}$$

$$1 - i = 1 - \frac{1}{n}$$



$$i = \frac{1}{2}$$

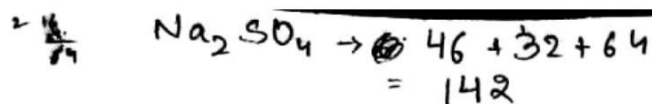
AIRMT-II

Ques The F.P. depression const. for water is $-1.86^\circ\text{Cm}^{-1}$. If 5.00 g Na_2SO_4 is dissolved in 45.0 g H_2O . The F.P. is changed by -3.82°C . Calculate the van't Hoff factor for Na_2SO_4 .

$$\Delta T_f = i K_f m$$

$$\begin{array}{r} 273 \\ - 3.82 \\ \hline 269.18 \end{array}$$

$$\begin{array}{r} 273 \\ - 3.82 \\ \hline 269.18 \end{array}$$



$$\cancel{269.18} + 3.82 = i \times (1.86) \times \frac{5}{142} \times 1000$$

$$\frac{186}{372}$$

$$\frac{382 \times 3 \times 71}{31 \times 1000} = i$$

$$\frac{93}{31} \times \frac{186 \times 5 \times 1000}{1000}$$

$$i = \frac{382 \times 3 \times 71}{31 \times 1000}$$

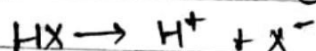
$$= \frac{381 \times 3 \times 71}{31 \times 1000}$$

$$\approx 2.7051$$

Ques 0.5 molal aq. soln of a weak acid HX is 20% ionised.

If K_f for water is 1.86 kg mol^{-1} The lowering in F.P. of soln is -

- ① -1.12 K ② 0.56 K ③ 1.12 K ④ -0.56 K



$$\alpha = \frac{i-1}{n-1} = 0.20 = \frac{i-1}{2-1}$$

$$\Delta T_f = i \times K_f \times m$$

$$= 1.20 \times 1.86 \times 0.5$$

$$= 0.600 \times 1.86$$

$$\Delta T_f = 1.116^\circ \text{C}$$

$$0.20 = \frac{i-1}{1}$$

$$0.20 + 1 = i$$

$$i = 1.20$$

Ques A 0.0020 m aq. soln of an ionic compd. $Co(NH_3)_5$

(NO_2) ll Freeze at -0.00732°C . No. of moles of ion which 1 mol of ionic compd. produces on being dissolved in water will be ($K_f = -1.86^\circ \text{C/m}$)

- ① 2 ② 3 ③ 4 ④ 1

$$\Delta T$$

Ques 2.925 g of NaCl is dissolved in 100 g water. Determine B.P. of solⁿ (K_b for water = $0.52 \text{ K kg mol}^{-1}$)

$$\Delta T_b = K_b m$$

$$T_b - 373 = 0.52 \times \frac{2.925}{58.5} \times 1000$$

$$T_b = 0.52 \times \frac{2.925}{100 \times 58.5} \times 1000$$

$$= \frac{52 \times 5 \times 10}{10000} = 0.2$$

Calc. % DOD of ~~an~~ electrolyte XY_2 (Normal $M_w = 164$) in water. If observed M_w by measuring elevation in bp is 0.56.

$$i = 0.56$$

$$\alpha = \frac{i-1}{n-1}$$

$$n = 1 + 2 = 3$$

$$\alpha \% = \frac{0.56 - 1}{3 - 1} \times 100\%$$

=

Ques F.P. of 0.01 M aq. solⁿ of $K_3[Fe(CN)_6]$ is -0.062°C . What is α for this complex. If K_f for water = $1.26 \text{ K kg mol}^{-1}$)

$$\Delta T_b = i K_b m$$

$$0 - (-0.062) = i \times 1.26 \times 0.01$$

$$\frac{0.062}{1.26 \times 0.01} = i$$

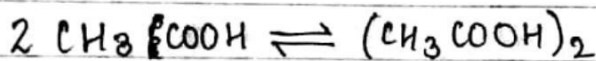
$$i = 4.9 \approx 5$$

$$\alpha = \frac{i - 1}{n - 1}$$

$$\alpha = \frac{5 - 1}{4 - 1} = \frac{4}{3}$$

Que 1 molal carboxylic acid in benzene shows a elevation in B.P. of 1.518 K . Calculate Degree of dissociation for dimerisation of acid in benzene K_b for benzene $= 2.53 \text{ K Kg mol}^{-1}$

$$\Delta T_b = i K_b m \quad \Rightarrow \quad i = \frac{\Delta T_b}{K_b m} = \frac{1.518}{2.53 \times 1} = 0.6$$



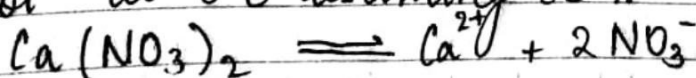
$$\begin{array}{ccc} 1 & & 0 \\ 1 - \alpha & & \alpha/2 \\ i = \frac{1 - \alpha/2}{1} \end{array}$$

$$0.6 = 1 - \frac{\alpha}{2}$$

$$\frac{\alpha}{2} = 1 - 0.6 = 0.4$$

$$\alpha = 0.8$$

Que Calculate the osmotic pr. of 20% w/v anhy. $\text{Ca}(\text{NO}_3)_2$ solⁿ at 0°C assuming 80% dissociation.



$$\begin{array}{ccc} 1 & & 2 \\ 1 - \alpha & & \alpha \quad 2\alpha \end{array}$$

$$i = \frac{1 - \alpha + \alpha + 2\alpha}{1} = \frac{1 + 2\alpha}{1} = 1 + 2 \times 0.8 = 1 + 1.6 = 2.6$$

$$\begin{aligned} \text{Ca(NO}_3)_2 &= 40 + 28 + 6 \times 16 \\ &= 68 + 96 \\ &= 164 \end{aligned}$$

39

$$\begin{aligned} \pi &= iCRT \\ &= \frac{W}{M} \frac{RT}{V} \times i \\ &= \frac{20}{164} \times \frac{1000}{1000} \times \frac{0.082}{1000} \times 273 \times 2.6 \\ \pi &= 27.30 \times 2.6 \\ &= \end{aligned}$$

20 gm solute + nt
in 100 ml soln

Que 2 solⁿ of glucose have osmotic pr 1.5 atm & 2.5 atm resp. 1 L of 1st solⁿ is mixed with 2 L of 2nd solⁿ. Find osmotic Pr. of resultant solⁿ.

$$\begin{aligned} \pi &= iCRT \\ &= i \frac{n}{V} RT \end{aligned}$$

$$\pi = i \frac{n}{V} RT$$

$$\frac{\pi V}{RT} = in$$

$$\begin{aligned} \pi V_T &= (in)_{\text{Total}} RT \\ &= \left(\frac{\pi_1 V_1}{RT} + \frac{\pi_2 V_2}{RT} \right) RT \end{aligned}$$

$$\begin{aligned} \pi V_T &= \pi_1 V_1 + \pi_2 V_2 \\ &= 1.5 \times 1 + 2.5 \times 2 \end{aligned}$$

$$\begin{aligned} \pi \times 3 &= 1.5 + 5.0 = 6.5 \\ \pi &= \frac{6.5}{3} = 2.1 \end{aligned}$$

Que Following solⁿ are given arrange them in ↑ order of V.P., B.P., F.P. & D.P. solⁿ.

① 1 M sugar $i=1$ ② 1 M Na₂SO₄ $i=3$ ③ 1 M KCl $i=2$ ④ 1 M Al₂(SO₄)₃ $i=5$

V.P. → 4 < 2 < 3 < 1

B.P. → 1 < 3 < 2 < 4

F.P. → 4 < 2 < 3 < 1

O.P. → 1 < 3 < 2 < 4

$$\Delta T_b = T_b - T_b^\circ$$

$$\Delta T_b + T_b^\circ = T_b$$

B.P. ↑

$$\Delta T_f = T_f^\circ - T_f$$

$$T_f = T_f^\circ - \Delta T_f$$

F.P. ↓

$$P^\circ - P_T = \Delta P$$

$$P^\circ - \Delta P = P_T$$

V.P. ↓

$$\pi = iCRT$$

Ques Find maxⁿ & minⁿ c.p. of following aq. solⁿ.

① 1% W/V sugar	② 1% W/V urea	③ 1% W/V Na ₂ SO ₄	④ 1% W/V NaCl
$C = \frac{1/342 \times 1000}{100}$	$C = \frac{1/60 \times 1000}{100}$	$C = \frac{1/142 \times 1000}{100}$	$C = \frac{1/58.5 \times 1000}{100}$
$C = \frac{10}{342}$	$C = \frac{10}{60}$	$C = \frac{10}{142}$	$C = \frac{10}{58.5}$
$i \times C = 0.029$ ↓ min ⁿ	$= \frac{1}{6} = 0.16$	$= 3 \times \frac{10}{142} = 0.211$	$= 0.34$ ↓ max

c.p. & observed concⁿ (i x c)

AIPMT-14

Ques Which of the following solⁿ will exhibit the largest F.P. depression?

- ① KCl ② C₆H₁₂O₆ ③ Al₂(SO₄)₃ ④ K₂SO₄
 $i = 2$ $i = 1$ $i = 5$ $K = 3$

$$\Delta T_{\text{f}} = i K_{\text{f}} m$$

Ques An aq. solⁿ is 1.00 m in KI which change will cause ↑ in v.p. of solⁿ.

- ① addⁿ of NaCl ② Addⁿ of Na₂SO₄ ③ Addⁿ 1.00 molal KI
 ④ addⁿ of water

Ques which of the following aq. solⁿ have min^m F.P.?

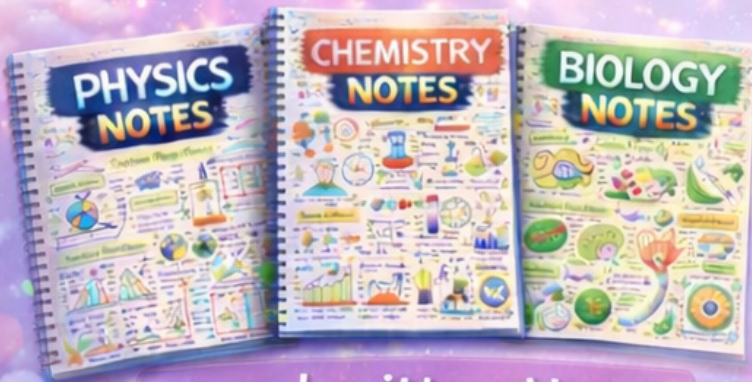
- ① 0.01 M NaCl $i=2$ $T_f = T_f^0 - 0.02 K_f$
 ② 0.005 m MgI_2 $i=3$ $T_f = T_f^0 - 0.015 K_f$
 ③ $0.005\text{ m C}_2\text{H}_5\text{OH}$ $i=2$ $T_f = T_f^0 - 0.010 K_f$
 ④ 0.005 m MgSO_4 $i=2$ $= T_f^0 - 0.010 K_f$

$$\Delta T_f = T_f^0 - T_f$$

$$T_f = T_f^0 - \Delta T_f$$

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